Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If x and a stand for distance, then for what value of n is the given equation dimensionally correct? The equation is $\int \frac{dx}{\sqrt{a^2-x^n}} = \sin^{-1}\frac{x}{a}$.

(A) Zero

(B) 2

(C) -2

(D) 1

Q.2

An experiment measures quantities a, b, c and then X is calculated from $X = \frac{a^{1/2}b^2}{c^3}$. If the percentage errors in a, b and c are $+1\% \pm 3\%$ and $\pm 2\%$ respectively, then the percentage error in X can be

(A) 12.5%

(B) 7%

(C) 1%

(D) 4%

Q.3

If P represents radiation pressure, c represents speed of light and Q represents radiation striking unit area per second, then non-zero integers x, y, and z such that $P^x Q^y c^z$ is dimensionless are

(A) $x = 1, y = 1, z = -1$

(B) $x = 1, y = -1, z = 1$

(C) $x = -1, y = 1, z = 1$

(D) $x = 1, y = 1, z = 1$

Q.4

The velocity of a body is given by the equation $v = \frac{b}{t} + ct^2 + dt^3$
The dimensional formula of b is

(A) $\left[M^0LT^0\right]$

(B) $\left[M^0T^0\right]$

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(C) $\left[M^0L^0T\right]$

(D) MLT^{-1}

Q.5

The length l, breadth b and thickness t of a block of wood are measured with the help of a meter scale. The results after calculating the errors are given as $l = 15.12 \pm 0.01$ cm, $b = 10.15 \pm 0.01$ cm $\quad t = 5.28 \pm 0.01$ cm. The percentage error in volume upto proper significant figures is

(A) 0.35%

(B) 0.28%

(C) 0.48%

(D) 0.64%

Q.6

In the formula $X = 3YZ^2$, X and Z have the dimensions of capacitance and magnetic induction, respectively. The dimensions of Y in M.KS system are

(A) $M^{-3}L^{-2}T^{-2}A^{-4}$

(B) $ML^{-2}A$

(C) $M^{-3}L^{-2}T^{2}A^{4}$

(D) $M^{-3}L^{-2}T^{4}A^{4}$

Q.7

In the formula, $a = 3bc^2$, a and c have dimensions of electric capacitance and magnetic field respectively. The dimensions of b in S.I. system are

(A) $\left[\mathrm{M}^{-3}\mathrm{L}^{-2}\mathrm{T}^{4}\mathrm{Q}^{4}\right]$

(B) $\left[\mathrm{M}^{-3}\mathrm{T}^{4}\mathrm{Q}^{4}\right]$

(C) $\left[\mathrm{M}^{-3}\mathrm{T}^{3}\mathrm{Q}\right]$

(D) $\left[\mathrm{M}^{-3}\mathrm{L}^{2}\mathrm{T}^{4}\mathrm{Q}^{-4}\right]$

Q.8

Frequency (n) of a tuning fork depends upon length (l) of its prongs, density (p) and Young's modulus (Y) of its material. Then frequency and Young' modulus will be related as

(A) $n \propto \sqrt{Y}$

(B) $n \propto Y$

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(C) $n \propto \frac{1}{\sqrt{Y}}$

(D) $n \propto \frac{1}{Y}$

Q.9

If L, C and R denote the inductance, capacitance and resistance respectively, the dimensional formula for C^2LR is

(A) $\left[\mathrm{ML^{-2}\,T^{-1}I^0}\right]$

(B) $\left[\mathrm{M^0L^0T^3}\right]$

(C) $\left[\mathrm{M^{-1}L^{-2}T^6I^2}\right]$

(D) $\left[\mathrm{M^0L^0T^2I^0}\right]$

Q.10

The volume V of water passing any point of a uniform tube during t seconds is related to the cross-sectional area A of the tube and velocity u of water by the relation

$$V \propto A^{\alpha}u^{\beta}t^{y}$$

which one of the following will be true?

(A) $\alpha = \beta = \gamma$

(B) $\alpha \neq \beta = \gamma$

(C) $\alpha = \beta \neq \gamma$

(D) $\alpha \neq \beta \neq \gamma$

Q.11

The ratio of the dimensions of Planck's constant and that of the moment of inertia is the dimensions of

(A) time

(B) frequency

(C) angular momentum

(D) velocity

Q.12

Two quantities A and B have same dimensions which mathematical operation given below is physically meaningful?

(A) A/B

(B) $\mathrm{A} + \mathrm{B}^2$

(C) $A^2 - B$

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(D) $A = B^2$

Q.13

The solar constant is defined as the energy incident per unit area per second. The dimensional formula for solar constant is

(A) $\left[M^0\ L^0\ T^0\right]$

(B) $\left[MLT^{-2}\right]$

(C) $\left[ML^2\ T^{-2}\right]$

(D) $\left[ML^0\ T^{-3}\right]$

Q.14

Distance travelled by a particle at any instant ' t ' can be represented as $S = A(t + B) + Ct^2$. The dimensions of B are

(A) $\left[M^0\ L^1\ T^{-1}\right]$

(B) $\left[M^0\ L^0\ T^1\right]$

(C) $\left[M^0\ L^{-1}\ T^{-2}\right]$

(D) $\left[M^0\ L^2\ T^{-2}\right]$

Q.15

If P, Q, R are physical quantities, having different dimensions, which of the following combinations can never be a meaningful quantity?

(A) $(P - Q)/R$

(B) $PQ - R$

(C) PQ/R

(D) $\left(PR - Q^2\right)/R$

Q.16

If speed V, force F and acceleration a are chosen as the fundamental physical quantities, then the dimension of Young's modulus in terms of V, F and a are

(A) $\left[V^{-3}Fa\right]$

(B) $\left[V^{-4}F^2a^2\right]$

(C) $\left[V^{-4}Fa^2\right]$

(D) $\left[V^{-4}F^2a\right]$

Q.17

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If $v = \frac{a}{t} + bt^3$ where $v =$ velocity and t is time The dimensional formula of a and bare

(A) $[\mathrm{T}], \left[\mathrm{T}^{-3}\right]$

(B) $[\mathrm{L}], \left[\mathrm{LT}^{-4}\right]$

(C) $\left[\mathrm{T}^{-3}\right], [\mathrm{T}]$

(D) $\left[\mathrm{LT}^{-4}\right], [\mathrm{L}]$

Q.18

What is the dimensional formula for Planck's constant and angular momentum?

(A) $\left[ML^2T^{-2}\right]$ and $\left[MLT^{-1}\right]$

(B) $\left[ML^2T^{-1}\right]$ and $\left[ML^2T^{-1}\right]$

(C) $\left[ML^3T^1\right]$ and $\left[ML^2T^{-2}\right]$

(D) $\left[MLT^{-1}\right]$ and $\left[MLT^{-2}\right]$

Q.19

In the equation $\mathrm{P} = \frac{\mathrm{RT}}{\mathrm{V}-\mathrm{b}}\mathrm{e}^{\frac{\mathrm{aV}}{\mathrm{RT}}}$
$\mathrm{V} =$ volume, $\mathrm{P} =$ pressure, $\mathrm{R} =$ universal gas constant, and $\mathrm{T} =$ temperature The dimensional formula of a is same as that of

(A) V

(B) P

(C) T

(D) R

Q.20

Let Q denote the charge on the plate of a capacitor of capacitance C. The dimensional formula for $\frac{\mathrm{Q}^2}{\mathrm{C}}$ is

(A) $\left[\mathrm{L}^2\mathrm{M}^2\ \mathrm{T}\right]$

(B) $\left[\mathrm{LMT}^2\right]$

(C) $\left[\mathrm{L}^2\mathrm{MT}^{-2}\right]$

(D) $\left[\mathrm{L}^2\mathrm{M}^2\ \mathrm{T}^2\right]$

Answers & Solutions

Q.1 Answer:

2

Solution:

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Here, x^n has the same dimensions as a^2. Thus, $n = 2$ will make the expression dimensionally correct.

Q.2 Answer:

12.5%

Solution:

$X = \frac{a^{1/2}b^2}{c^3}$

$\frac{\Delta x}{X} \times 100 = \frac{1}{2}\frac{\Delta a}{a} \times 100 + 2\frac{\Delta b}{h} \times 100 + 3\frac{\Delta c}{c} \times 100 = \frac{1}{2} \times 1 + 2 \times 3 + 3 \times 2 = 12.5\%$

Q.3 Answer:

$x = 1, y = -1, z = 1$

Solution:

Dimensions of

$P^xQ^yc^z = \left[ML^{-1}T^{-2}\right]^x \left[MT^{-3}\right]^\gamma \left[LT^{-1}\right]^z$

As it is dimensionless, so

$\left[ML^{-1}T^{-2}\right]^x \left[MT^{-3}\right]^y \left[LT^{-1}\right]^z = \left[M^0L^0T^0\right]$ **or** $\left[M^{x+y}L^{-x+2}T^{-2x-3y-z}\right] = \left[M^0L^0T^0\right]$

Comparing powers of M, L and T, we get

$x + y = 0, -x + z = 0, -2x - 3y - z = 0$

Solving, $x = 1, y = -1, z = 1$.

Q.4 Answer:

$\left[M^0LT^0\right]$

Solution:

In the equation, left hand side has the dimension of velocity. Thus, from the principle of homogeneity each term in the right hand side should have the dimensions of velocity.

$\left[\frac{b}{t}\right] = [v]_o[b] = [vt] = [L]$

Q.5 Answer:

0.35%

Solution:

$V = lbt$

$\frac{\Delta V}{V} \times 100 = \frac{\Delta}{l} \times 100 + \frac{\Delta b}{b} \times 100 + \frac{\Delta t}{t} \times 100$

$= \frac{0.01}{15.12} \times 100 + \frac{0.01}{10.15} \times 100 + \frac{0.01}{5.28} \times 100$

$= 0.35405\%$

Q.6 Answer:

$M^{-3}L^{-2}T^4A^4$

Solution:

The given relation is:

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$$X = 3YZ^2$$

$$Y = \frac{1}{3}\frac{X}{Z^2}$$

since X is the capacitance. So, its dimensions are: $X = M^{-1}\,L^{-2}\,T^2Q^2$
and Z is the magnetic induction and its dimensions are $Z = MT^{-1}Q^{-1}$
Therefore, Y can be expressed as:

$$Y = \frac{M^{-1}\,L^{-2}\,T^2Q^2}{\left[MT^{-1}Q^{-1}\right]^2} = M^{-3}\,L^{-2}\,T^4Q^4$$

Q.7 Answer:

$[M^{-3}L^{-2}T^4Q^4]$

Solution:

Given, $a = 3bc^2$

$\Rightarrow \quad b = \frac{a}{3c^2}$

Writing dimensions for a and c, we have

$$b = \frac{Q/V}{B^2} = \frac{Q\left[ML^2T^{-2}Q^{-1}\right]^{-1}}{MT^{-1}Q^{-1^2}}$$

$= [M^{-3}L^{-2}T^4Q^4]$

Q.8 Answer:

$n \propto \sqrt{Y}$

Solution:

Let $n \propto l^a \rho^b Y^c$

Putting dimensions of all the quantities, we have $\left(T^{-1}\right) \propto L^a \left(ML^{-3}\right)^b \left(ML^{-1}T^{-2}\right)^c$

Equating powers of M, L **and** T **on both sides, we get** $b + c = 0, a - 3b - c = 0$ **and** $-2c = -1$ **which give** $a = -1, b = -\frac{1}{2}$ **and** $c = \frac{1}{2}$**. Thus** $n \propto l^{-1}\rho^{1/2}Y^{1/2}$.

Q.9 Answer:

$M^0L^0T^3$

Solution:

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$C^2LR = C^2L^2 \times \frac{R}{L}$

As $\frac{L}{R}$ and $\sqrt{LC}$ both have the dimensions of time, we get

$C^2LR = \mathrm{M^0L^0T^4} \times \mathrm{M^0L^0T^{-1}} = \mathrm{M^0L^0T^3}$

Q.10 Answer:

$\alpha \neq \beta = \gamma$

Solution:

The dimension of the two sides of proportionality are

$L^3 = L^{2\alpha}\left(LT^{-1}\right)^{\beta} T^{y} = L^{2n+\beta}T^{\gamma-\beta}$

Equating the powers of dimensions on both sides, we have

$2\alpha + \beta = 3$

$\gamma - \boldsymbol{\beta} = \mathbf{0}$

which give $\beta = \gamma$ and $\alpha = \frac{1}{2}(3-\beta)$

i.e., $\alpha \neq \beta = \gamma$.

Q.11 Answer:

frequency

Solution:

$\frac{\text{Planck's constant}}{\text{Moment of inertia}} = \frac{\frac{2\pi \mathrm{I}\omega}{\mathrm{n}}}{\mathrm{I}}$

[As $\frac{\mathrm{nh}}{2\pi} = \mathrm{I}\omega$**]**

$$= \frac{2\pi \mathrm{I}(2\pi \mathrm{f})}{\mathrm{nI}} = \left(\frac{4\pi^2}{\mathrm{n}} \cdot \mathrm{f}\right) = \left[\mathrm{T}^{-1}\right]$$

Q.12 Answer:

A/B

Solution:

$[\mathrm{A}] \neq \left[\mathrm{B}^2\right]$ **so, cannot be added or subtracted**

Q.13 Answer:

$\left[\mathrm{ML}^0\ \mathrm{T}^{-3}\right]$

Solution:

Energy incident per unit area per second $= \frac{\text{Energy}}{\text{area} \times \text{second}} = \frac{\mathrm{ML}^2\ \mathrm{T}^{-2}}{\mathrm{L}^2\ \mathrm{T}} = \mathrm{MT}^{-3}$

Q.14 Answer:

$\left[\mathrm{M}^0\ \mathrm{L}^0\ \mathrm{T}^1\right]$

Solution:

In $\mathrm{S} = \mathrm{A}(\mathrm{t} + \mathrm{B}) + \mathrm{Ct}^2$; **B is added to time t.**

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Therefore, dimensions of B are those of time.

Q.15 Answer:

$(P - Q)/R$

Solution:

$[P] \neq [Q]$ so, $P - Q$ **is not meaningful**

Q.16 Answer:

$V^{-4}Fa^2$

Solution:

Let $Y = fV, F, a$

$\therefore Y = KV^x F^y a^z$, $K \rightarrow$ **Unitless**

$\therefore Y = V^x F^y a^z$

$\therefore ML^{-1}T^{-2} = LT^{-1}{}^x MLT^{-2}{}^y LT^{-2}{}^z$

$\therefore ML^{-1} = T^{-2} = M^y L^{x+y+z} T^{-x-2y-2z}$

$\therefore y = 1; \quad x + y + z = -1; \quad -x - 2y - 2z = 2$

$\therefore x + z = -2 \quad \therefore x + 2y + 2z = 2$

$\therefore x + 2z = 0 \quad \therefore z = 2, x = -4$

$\therefore Y = V^{-4}Fa^2$

Q.17 Answer:

$[L], [LT^{-4}]$

Solution:

Dimensionally $v = \frac{a}{t} \Rightarrow a = LT^{-1} \times T = L$

also dimensionally $v = bt^3 \Rightarrow b = \frac{LT^{-1}}{T^3} = LT^{-4}$

Q.18 Answer:

ML^2T^{-1} and ML^2T^{-1}

Solution:

Angular momentum of a particle moving in circular path about centre is

$L_o = mvR$

By principle of homogeneity

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$L_o = mvR = M^1 \times L^1 T^{-1} \times L^1$

i.e. $L_o = M^1 L^2 T^{-1}$

Now, according quantization of bohr orbits, angular momentum of an electron is given as $L_o = \frac{nh}{2\pi}$

By principle of homogeneity

$L_o = h$

Because $\frac{n}{2\pi}$ **is dimensionless.**

Q.19 Answer:

P

Solution:

$$[\mathrm{a}] = \frac{[\mathrm{RT}]}{[\mathrm{V}]} = [\mathrm{P}]$$

Q.20 Answer:

$\left[\mathrm{L}^2\mathrm{MT}^{-2}\right]$

Solution:

We know that $\frac{\mathrm{Q}^2}{2\mathrm{C}}$ **is energy of capacitor, so it represent the dimension of energy** $= \left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right]$

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